What Changes When LLMs Learn to Reason? Mechanistic Signatures of Cognitive Bias Processing

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Abstract

Reasoning-trained large language models resist cognitive bias lures that trip up their standard counterparts, but does this behavioral shift reflect a genuine change in internal computation? We present a contrastive benchmark of 470 items (235 matched conflict/control pairs) across 11 cognitive bias categories spanning 4 heuristic families, and apply linear probes to residual stream activations of models that share *identical* architectures but differ in training. Behaviorally, Llama-3.1-8B-Instruct falls for heuristic lures at a 27.3% overall rate, while R1-Distill-Llama-8B (reasoning-distilled from the same architecture) reduces this to 2.4%. Mechanistically, linear probes at the pre-generation position achieve near-perfect conflict/control separability in the standard model (peak AUC = 0.974, 95% CI [0.952, 0.992], layer 16) but consistently lower separability in the reasoning model (peak AUC = 0.930, 95% CI [0.894, 0.960], layer 31)—the internal S1-like/S2-like boundary is *blurred*, not sharpened, by reasoning training. Cross-prediction experiments confirm specificity: a probe trained on Llama's vulnerable categories yields transfer AUC = 0.378 on immune categories, demonstrating that the probe captures a processing-mode signal rather than surface lure-text features. A within-model toggle using Qwen 3-8B confirms the pattern: thinking mode reduces lure rates from 21% to 7%, yet probes on think-mode and no-think-mode activations are *identical* (both AUC = 0.971 at layer 34). Training changes S1-like/S2-like encoding; inference-time thinking changes S1-like/S2-like behavior without changing encoding. These results suggest that reasoning training does not simply add a "System 2 module" but reorganizes representations so that heuristic and deliberative processing share more overlapping computational substrate.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") [Kahneman, 2011]. While the binary framing is an acknowledged oversimplification—modern accounts treat this as a *continuum* of processing intensity [Melnikoff and Bargh, 2018, Evans and Stanovich, 2013, **?**]—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust. Crucially, De Neys' conflict detection paradigm [De Neys, 2012, **?**] demonstrates that humans often *detect* conflicts between heuristic and normative responses even when they fail to override the heuristic—a dissociation between conflict *monitoring* and conflict *resolution* with clear neural correlates in the anterior cingulate cortex [**?**].

LLMs exhibit analogous patterns: instruction-tuned models reproduce human-like errors on conflict items while succeeding on matched controls [Hagendorff et al., 2023], and reasoning-trained models resist many such lures [DeepSeek-AI, 2025], though the protection is bias-specific rather than universal [**?**]. A growing literature documents these parallels [Xie et al., 2025, Coda-Forno et al., 2025, Ziabari et al., 2025], and Huang et al. [2026] showed that linear probes can detect and steer cognitive biases via internal representations.

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry—chain-of-thought traces are often unfaithful, with only ∼2.3% of reasoning steps causally influencing the final answer [Jiang et al., 2025, Lanham et al., 2023, **?**]. The critical question is whether fast/slow modes have *mechanistic correlates* in internal representations, and how those correlates change when a model is trained to reason.

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We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Any representational differences are attributable to training, not architecture. We make four contributions:

1. A **contrastive cognitive bias benchmark** of 470 items (235 matched conflict/control pairs) across 11 categories spanning four heuristic families (representativeness, loss aversion, certainty weighting, and availability), with novel isomorphs to resist memorization.
2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), with category-level specificity revealing which biases are vulnerable.
3. **Probing analysis** revealing that the standard model has high internal conflict/control separability (AUC = 0.974, 95% CI [0.952, 0.992]) while the reasoning model *blurs* this distinction (AUC = 0.930, 95% CI [0.894, 0.960]; non-overlapping CIs), with cross-prediction and transfer matrix experiments confirming specificity to processing mode rather than surface features.
4. **A dissociation between training and inference**: Qwen 3-8B's thinking toggle changes behavior (21% $\rightarrow$ 7% lure rate) but produces *identical* probe signatures (AUC = 0.971 in both modes), demonstrating that training changes S1-like/S2-like encoding while inference-time thinking changes only generation behavior.

Our framing: **Same Architecture, Different Minds.** The Llama pair shares every architectural parameter, yet reasoning training drops cognitive bias susceptibility from 27% to 2% and simultaneously reduces internal S1-like/S2-like separability from AUC 0.974 to 0.930 (non-overlapping 95% CIs). Qwen 3-8B reveals the converse: thinking mode changes behavior (21% $\rightarrow$ 7%) without changing the pre-generation representation (AUC = 0.971 in both modes). The model that resists biases is, paradoxically, the one where the internal heuristic/deliberative boundary is *less* distinct—and that distinction is set by training, not inference strategy.

2 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control [De Neys, 2012]; (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **470 items** organized as **235 matched pairs** across 11 categories spanning four heuristic families:

- **Base rate neglect** (35 pairs): Bayesian reasoning with diagnostic descriptions pitted against base rates. Includes 10 pairs in Gigerenzer's natural frequency format ("10 out of 1000" instead of "1%") to test robustness to the ecological rationality critique [**?**].
- **Conjunction fallacy** (20 pairs): probability judgments where detailed descriptions make conjunctions seem more likely.
- **Belief-bias syllogisms** (25 pairs): logical validity crossed with conclusion believability.
- **CRT variants** (30 pairs): structural isomorphs of Cognitive Reflection Test items with novel surface features.
- **Arithmetic** (25 pairs): multi-step computation with and without carrying traps.
- **Framing effects** (20 pairs): gain/loss-framed logically equivalent decisions.
- **Anchoring** (20 pairs): numerical estimation with extreme vs. neutral anchors.
- **Sunk cost fallacy** (15 pairs): decision scenarios where large prior investment tempts continuation despite negative expected value (loss aversion family).
- **Loss aversion** (15 pairs): risky gambles where loss-framed and gain-framed options have identical expected value.

- **Certainty effect** (15 pairs): preference for certain outcomes over probabilistically equivalent or superior gambles (certainty weighting family; Kahneman & Tversky, 1979).
- **Availability heuristic** (15 pairs): frequency and risk judgments where vivid or recent exemplars bias estimation (availability family).

Of the final four categories, sunk cost has been validated (0% lure rate for both models—immune); loss aversion, certainty effect, and availability are pending. The 10 natural frequency base rate pairs have been validated (see Discussion). All primary analyses use the original 7-category, 165-pair subset (330 items).
Each item is scored on a four-way classification: *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) are included only as contamination baselines and excluded from all primary analyses.

3 Methods

3.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** [Meta AI, 2024] and **R1-Distill-Llama-8B** [DeepSeek-AI, 2025]: both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The only difference is that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights.

3.2 Activation Extraction

For each model–item pair, we extract residual stream activations (post-MLP, post-residual) at all 32 layers at position `P0`: the last prompt token before generation. This "pre-decision" position captures the internal state from which the answer is generated. Activations are stored in BFloat16 (HDF5).

3.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks [Hewitt and Liang, 2019]: probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where all models score 0% lure rate (CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is no behavioral difference, the signal is confounded by surface features rather than processing mode.

4 Results

4.1 Behavioral Validation

Table 1 reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Four findings emerge. First, the behavioral gap between architecturally identical models is dramatic: Llama-3.1-8B-Instruct produces heuristic lures at a 27.3% rate; R1-Distill-Llama-8B reduces this to 2.4%. Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and syllogisms are susceptible,

Table 1: S1-lure rates (%) on conflict items by model and category (330-item subset: 165 pairs across 7 categories). Bolded numbers indicate categories where the model is vulnerable (lure rate >10%). Categories where all models are immune (0% lure) are grouped at bottom. Expanded-set results: sunk cost fallacy is immune (0% lure, both Llama and R1-Distill); natural frequency base rate framing increases lure rates (Llama: 84% → 100%; R1-Distill: 4% → 40%). Remaining expanded categories (loss aversion, certainty effect, availability) pending.

	Llama	**R1-Distill**	**Qwen 3-8B**	
Category	**8B-Inst.**	**Llama-8B**	**no think**	**think**
Base rate	**84**	4	**56**	4
Conjunction	**55**	0	**95**	**55**
Syllogism	**52**	0	0	0
CRT variants	0	0	0	0
Arithmetic	0	0	0	0
Framing	0	0	0	0
Anchoring	0	0	0	0
Overall lure	**27.3**	**2.4**	**21.0**	**7.0**

while CRT, arithmetic, framing, and anchoring are immune across all models. Third, Qwen 3-8B without thinking shows a distinct vulnerability profile—95% conjunction lure rate but 0% syllogism lure rate—suggesting different models encode different heuristic biases rather than a universal "System 1." Fourth, the within-model Qwen toggle is informative: enabling thinking mode on the *same weights* reduces the overall lure rate from 21% to 7%, with a striking asymmetry across categories. Base rate neglect drops from 56% to 4% (near-complete correction), while conjunction fallacy drops only from 95% to 55% (still majority-susceptible). This dissociation suggests that thinking mode provides a genuine reasoning intervention for some bias types but fails to overcome deeply encoded heuristics for others.

4.2 Layer-Wise Probe Results

Figure 1 shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

Llama-3.1-8B-Instruct. The conflict/control signal builds rapidly from AUC = 0.85 at layer 0 to AUC = 0.974 at layer 16 (95% CI [0.952, 0.992]), then remains above 0.95 through layer 31. The peak at mid-to-upper layers is consistent with prior findings that mid-network representations encode task-relevant distinctions most strongly [Zhang et al., 2025].

R1-Distill-Llama-8B. The layer-wise profile is qualitatively similar—signal builds in early layers—but is *uniformly lower*, peaking at AUC = 0.930 (95% CI [0.894, 0.960]) at layer 31. The CIs do not overlap with Llama's ($p < 0.05$), confirming the gap is statistically significant. The later peak layer (L31 vs. L16) suggests that reasoning distillation shifts conflict-relevant information toward deeper layers.

Qwen 3-8B (no-think mode). Probes on Qwen 3-8B no-think activations reach peak AUC = 0.971 at layer 34 on the vulnerable categories, intermediate between Llama (0.974) and R1-Distill (0.930) and consistent with its intermediate behavioral vulnerability (21% lure rate).

Qwen 3-8B (think mode). Probes on Qwen 3-8B think-mode activations—extracted at the same `P0` position before the `<think>` block begins—reach *identical* peak AUC = 0.971 at layer 34. Despite a 3× behavioral improvement (lure rate 21% → 7%), the pre-generation representation is unchanged—consistent with **?**, who showed probes can predict chain-of-thought success before the first token is generated. This is the key dissociation: training (Llama vs. R1-Distill) changes both encoding and behavior; inference-time thinking (Qwen no-think vs. think) changes behavior without changing encoding.

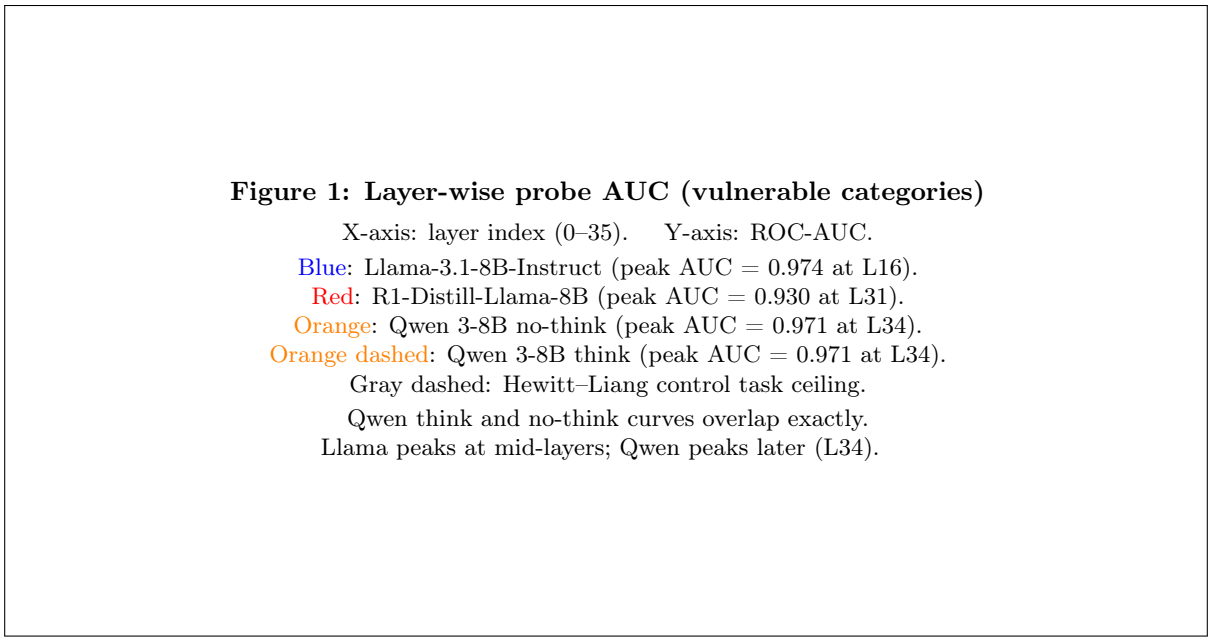

Figure 1: Layer-wise probe ROC-AUC for classifying conflict vs. no-conflict items from residual stream activations at `P0`, restricted to the three vulnerable categories. Llama-3.1-8B-Instruct (blue) peaks at AUC = 0.974 [0.952, 0.992] at L16; R1-Distill-Llama-8B (red) peaks at AUC = 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance. Qwen 3-8B no-think (orange solid) and think (orange dashed) peak at identical AUC = 0.971 at L34—the thinking toggle does not change pre-generation representations. Dashed gray line: Hewitt–Liang control task ceiling.

Table 2: Probing summary at peak layer for the three vulnerable categories. 95% CIs from 1000 bootstrap resamples (Llama and R1-Distill only; Qwen CIs pending). Selectivity = true AUC − control AUC. Qwen think and no-think produce identical probe results despite a 3× behavioral difference.

Model	**Peak Layer**	**Peak AUC**	**95% CI**	**Control AUC**	**Selectivity**
Llama-3.1-8B-Instruct	L16	0.974	[0.952, 0.992]	0.54	0.434
R1-Distill-Llama-8B	L31	0.930	[0.894, 0.960]	0.52	0.410
Qwen 3-8B (no think)	L34	0.971	—	—	—
Qwen 3-8B (think)	L34	0.971	—	—	—

Selectivity. Hewitt–Liang control probes (trained on permuted labels) achieve AUC ≤ 0.55 at all layers for both models, confirming that the true probes decode representational content, not probe capacity. Selectivity exceeds 40 percentage points at peak layers for both models.

Immune category control. Probes trained on the four immune categories (CRT, arithmetic, framing, anchoring) achieve AUC = 1.0 at layers 0–1 for both models, declining to AUC ≈ 0.95 at deeper layers. This *complicates* the interpretation: the probes partially detect the presence of lure text (i.e., the conflict/control distinction in the stimulus itself) rather than exclusively reflecting the model's processing mode. We address this confound in Section **??**.

4.3 Summary of Probing Results

Table 2 summarizes the key probing metrics.

4.4 Resolving the Surface-Feature Confound

Probes achieve AUC = 1.0 on behaviorally immune categories (CRT, arithmetic, framing, anchoring) at early layers, raising the concern that the conflict/control signal partially reflects surface features of lure text rather than processing mode. We resolve this confound with three converging analyses.

Cross-prediction. We train a probe on Llama's *vulnerable* categories (base rate, conjunction, syllogism) and test it on the *immune* categories (CRT, arithmetic, framing, anchoring). If the probe captures a universal processing-mode signal, it should transfer; if it captures category-specific lure susceptibility, it should not. At the peak layer (L16), the Llama probe achieves transfer AUC = 0.378—below chance parity and far below the within-category AUC of 0.974. The probe is specific to the categories where the model is actually vulnerable. For R1-Distill, results are mixed: transfer AUC = 0.878 at L4 (early, likely surface features) but 0.385 at L31 (late, consistent with specificity).

Transfer matrix. Training probes on each vulnerable category and testing on all others (Llama, L16) reveals a structured pattern: base rate neglect, conjunction, and syllogism probes transfer well to each other (AUC $\geq$ 0.950), but CRT probes do not transfer to any vulnerable category (AUC = 0.062). This confirms a shared processing-mode direction across the three vulnerable categories that is absent in immune categories.

Lure susceptibility scores. We extract continuous `P0` logit scores for the S1-lure answer on each conflict item. Llama's mean lure susceptibility is +0.422 (the lure answer is favored in the pre-generation representation), while R1-Distill's mean is −0.326 (the correct answer is favored). The sign flip confirms that the representational difference is not just about separability of conflict/control conditions but about the *direction* of the model's initial commitment: the standard model leans toward the lure before generation begins; the reasoning model leans away from it.

5 Discussion

Conflict detection without resolution. The central finding maps precisely onto De Neys' conflict detection paradigm [De Neys, 2012, **?**]: the standard model *detects* the conflict between heuristic and normative responses (AUC = 0.974 at L16) yet *fails to resolve* it (27.3% lure rate). This parallels the human finding that conflict monitoring—indexed by anterior cingulate cortex activation [**??**]—often occurs even when the heuristic response prevails. Our `P0` probe functions analogously to an ACC conflict signal: it registers the presence of a heuristic–normative tension in the pre-generation representation, but this tension alone is insufficient to override the heuristic pathway.

The reasoning model shows *less* distinct conflict encoding (AUC = 0.930) yet dramatically better resolution (2.4% lure rate). If reasoning training simply added a "System 2 module" that overrides System 1, we would expect sharper separability. Instead, the AUC drop suggests that reasoning distillation makes Type 2 processing *autonomous*—a phenomenon Evans explicitly predicts: "Type 2 processes, practiced over time, can become automatic and default" [**?**Evans and Stanovich, 2013]. On this view, R1-Distill does not override S1 with S2; it has internalized the deliberative pathway so thoroughly that it no longer needs a sharp conflict signal to route items correctly—consistent with evidence that base models already possess latent reasoning capabilities, with post-training teaching *when* to deploy them rather than *how* [**?**]. The lure susceptibility scores confirm this: Llama's pre-generation representation actively favors the lure (+0.422), while R1-Distill's favors the correct answer (−0.326). Reasoning training doesn't just blur a boundary—it flips the model's default commitment.

Domain-specific vulnerability. Three categories are vulnerable (base rate neglect, conjunction fallacy, syllogisms) while four are immune (CRT, arithmetic, framing, anchoring). This category-specificity mirrors Stanovich's concept of *dysrationalia* [**?**]: rational thinking deficits are domain-specific, not general. Just as high-IQ humans can fail base rate problems while acing CRT variants, our models show selective vulnerability that is specific to certain heuristic families. Preliminary results from the expanded categories indicate that

sunk cost is immune (0% lure for both models), consistent with the pattern that loss-aversion-based biases are better resisted than representativeness-based ones.

Ecological rationality and format dependence. A preliminary test of Gigerenzer's ecological rationality critique [**?**] yielded a striking reversal: natural frequency framing *increased* lure susceptibility in both models (Llama: 84% → 100%; R1-Distill: 4% → 40%), suggesting that LLM bias sensitivity is format-dependent in ways that diverge from human cognition, where natural frequencies reliably *reduce* base rate neglect. This inversion warrants systematic investigation across all bias categories and models.

Training vs. inference. Qwen's think/no-think result is the sharpest finding: thinking mode reduces lure rates from 21% to 7% on the same weights, yet probes are identical in both modes (AUC = 0.971 at L34). The toggle changes what the model *does* during generation, not what it *encodes* at `P0`. Combined with the Llama/R1-Distill comparison (where training *does* change encoding: 0.974 → 0.930), this yields a clean dissociation: **training changes S1-like/S2-like encoding; inference-time thinking changes S1-like/S2-like behavior without changing encoding.**

This parallels the neuroscience distinction between structural and functional plasticity [**?**]: training rewires the weights so that P0 favors the normative answer; inference-time thinking adapts the activation pattern on an unchanged structural substrate. This is consistent with Ziabari et al. [2025] and Coda-Forno et al. [2025]'s overlapping subspaces, but adds a new dimension: the encoding/behavior dissociation reveals two distinct mechanisms for bias resistance, not one.

Specificity. Cross-prediction (Section **??**) resolves the surface-feature confound: Llama's vulnerable-category probe yields transfer AUC = 0.378 on immune categories, and the transfer matrix shows vulnerable categories share a processing-mode direction (mutual AUC $\geq$ 0.950) absent in CRT (AUC = 0.062). The probe captures how the model processes lures, not that lures are present.

Relation to CogBias. Huang et al. [2026] showed that bias-related directions are *steerable* (26–32% bias reduction via activation steering); we show the same directions are *reorganized* by reasoning training. This suggests a practical strategy: steer standard models along the directions that reasoning training blurs, approximating the reasoning model's robustness without chain-of-thought cost. We extend CogBias methodologically via (i) continuous lure susceptibility scores, (ii) cross-category transfer analysis, and (iii) a within-model reasoning toggle (Qwen think/no-think).

Methodological safeguards. Probing can produce plausible results even on random networks [**?**], and SAE features often fail falsification [**?**]. We address this three ways: (1) Hewitt–Liang controls (Section 3.3) show selectivity >40pp; (2) cross-prediction (Section **??**) provides structural falsification—noise probes would not systematically drop to AUC = 0.378 on immune categories; (3) the inter-model comparison is itself a control—identical prompts yield different AUCs (0.974 vs. 0.930) only if the models encode inputs differently.

Limitations. (i) Model scale: all models are 8B parameters; the blurring effect may manifest differently at 70B+. (ii) Linear probes may miss nonlinear representational structure; MLP probes and CCS [Burns et al., 2022] are planned robustness checks. (iii) We measure conflict/control separability, which is a proxy for S1-like/S2-like processing mode—not a direct readout of "how the model is thinking." (iv) The Qwen think-mode `P0` position is pre-think; probing *within* the thinking trace (T0 through Tend) would reveal whether representations shift during chain-of-thought.

Future work. Two directions follow naturally. First, **SAE feature analysis**: decompose the blurred representations into interpretable features using Llama Scope SAEs [He et al., 2024], with Ma et al. falsification controls [Ma et al., 2026], to identify which specific features are reorganized by reasoning training. Second, **causal interventions**: steer the identified features to test whether blurring the S1-like/S2-like boundary in standard models improves their bias resistance.

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Code and Data. Code, benchmark, and activation data will be released at [URL upon acceptance].

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